Final Year Project Proposal

AI-Powered Interview Preparation Assistant

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1. Introduction

Job seekers frequently struggle to prepare effectively for interviews due to the lack of personalized, on-demand practice and constructive feedback. Traditional interview preparation methods rely on generic question banks, self-study, or costly mock interview sessions with career coaches, which are not scalable or accessible to all students. This project proposes the development of a simple, easy-to-use desktop application that leverages Natural Language Processing (NLP) and Large Language Models (LLMs) to generate role-specific interview questions, accept the user's answers, and provide instant, AI-generated feedback — helping candidates prepare more effectively and confidently for real interviews.

2. Problem Statement

Students and job seekers lack access to affordable, personalized, and on-demand interview practice tools that simulate real interview scenarios and provide meaningful feedback on the content and clarity of their answers. There is a need for a simple, intelligent application that can generate relevant interview questions for a chosen job role and evaluate the user's answers automatically using AI-based techniques.

3. Objectives

- To design an application that generates role-specific technical and behavioral interview questions for a selected job domain.
- To allow users to type answers to the generated questions through a simple graphical interface.
- To evaluate submitted answers for relevance and clarity using NLP/LLM-based techniques.
- To generate short, constructive feedback and a score for each answer.

4. Scope

This project is scoped as a lightweight, single-user desktop application (not a web platform) to keep development simple and achievable within the project timeline. The application will provide a simple GUI where the user selects a job role/domain, receives generated interview questions one at a time, types an answer, and receives immediate AI-generated feedback and a score. The initial scope covers 2-3 common domains (e.g., Software Engineering and General HR/Behavioral) with text-based interaction only. Voice-based answers and web deployment are explicitly out of scope for this project.

5. Applications

- **University Career Services:** A mock interview practice tool students can use before campus placements.
- **Self-Paced Learning:** A personal practice tool for job seekers to rehearse independently, anytime.
- **Soft-Skills Development:** Feedback on answer clarity and completeness to improve communication skills.

6. Main Requirements (Feature Set)

6.1 Functional Requirements

Feature	Description
Job Role Selection	User selects a target job domain (e.g., Software Engineering, General HR/Behavioral) from a simple dropdown/menu in the GUI.
Interview Question Generation	Application generates a set of relevant technical/behavioral questions for the chosen domain using an LLM or a curated question bank.
Text-Based Answer Input	User types their answer to each question directly in the GUI.
Automated Answer Evaluation	Application evaluates the answer's relevance and completeness using NLP techniques (e.g., semantic similarity or LLM-based scoring).
Instant Feedback Display	Application displays short, constructive feedback and a score for each answer immediately after submission.
Session Summary	At the end of a practice session, the application shows an overall score for all questions attempted.

6.2 Non-Functional Requirements

- **Usability:** Simple, clean, and intuitive GUI requiring no technical expertise to operate.
- **Performance:** Question generation and answer evaluation should return results within a few seconds.
- **Reliability:** Consistent and accurate question generation and evaluation across repeated use.
- **Portability:** Runs as a standalone desktop application without requiring internet-dependent hosting (aside from optional LLM API calls).

7. Proposed Datasets

- **Interview Question Datasets (Kaggle):** Publicly available collections of common technical and behavioral interview questions across various domains, used to seed the base question bank.
- **Custom Curated Question Bank:** A small, manually curated set of questions for the 2-3 chosen domains (e.g., Software Engineering, HR/Behavioral) to supplement the public dataset.

8. Proposed Models and Tools

- **Question Generation:** Pretrained LLM (e.g., via API, or a small open-source model) prompted with the selected job domain, or selection from the curated question bank.
- **Answer Evaluation:** Sentence-BERT (SBERT) embeddings for semantic similarity scoring between the user's answer and reference key points, or direct LLM-based scoring.
- **GUI Development:** Python with Tkinter or PyQt for a simple, lightweight desktop interface.
- **NLP Libraries:** Hugging Face Transformers, spaCy, scikit-learn.

9. Evaluation Metrics

- Relevance and quality of AI-generated questions, assessed through informal review with sample users.
- Reasonableness of automated answer scores compared to the supervisor's/student's own judgment on a small set of test answers.

10. Project Timeline (As per University FYP Activity Chart)

Week	Date	Deliverable
Week 1	August 17, 2026	Meet faculty, decide project & advisor, fill FYP form with advisor's signature, submit to HOD.
Week 2	August 24, 2026	Project Proposal and Main Requirements (Feature Set) — this document.
Week 5	September 14, 2026	Requirements Specification (RS).
Week 8	October 5, 2026	R&D, Functional Spec (use case diagrams, conceptual GUI layout/navigation using Balsamiq), ERD, revised Project Plan.
Week 11	October 26, 2026	Design phase — architecture, class diagrams, sequence diagrams, prototype, and full database schema.
Week 13	November 9, 2026	Test Plan (Black-box testing, Usability testing).
Week 15	November 23, 2026	Complete RS and FS; working prototype in the actual environment (25-30% of RS implemented).
Week 16	November 23, 2026	Final presentation and submission of the final proposal.

11. Ethical Considerations

This application is intended strictly as a personal practice and self-improvement tool and is not designed to be used as an automated hiring or candidate-screening decision system. Any data entered by the user (answers) will be used only within the local application session and will not be shared with third parties.

12. Expected Outcome

A simple, functional desktop application with a graphical user interface that generates role-specific interview questions, accepts the user's typed answers, evaluates them using NLP/LLM techniques, and provides instant feedback and a score — enabling students to practice and improve their interview performance independently.